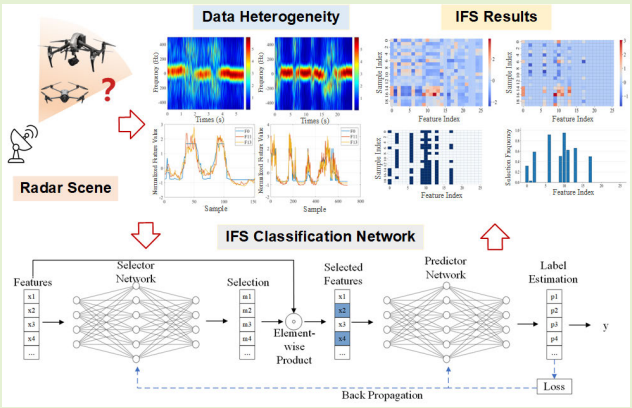
Instancewise Feature Selection for UAV Classification Based on Radar Micro-Doppler Signatures

Guidong He, *Graduate Student Member, IEEE*, Peng Lei, Jun Wang, and Jiangyou Zhu

***Abstract*— Unmanned aerial vehicle (UAV) classification using radar micro-Doppler signatures remains challenging due to the diversity, variability, and heterogeneity of the observed features. Most existing feature selection methods adopt a global strategy, which overlooks sample-specific differences arising from diverse UAV categories, flight dynamics, and observation conditions. To address this limitation, we propose an instancewise feature selection framework based on the information bottleneck (IB) principle. The method jointly optimizes a differentiable feature selector and a classifier via variational inference and the Gumbel–Softmax reparameterization, enabling adaptive identification of discriminative features for each sample. Experimental evaluations on real radar data validate the effectiveness and interpretability of the proposed approach.**



***Index Terms*— Information bottleneck (IB), instancewise feature selection, micro-Doppler signatures, radar, Unmanned aerial vehicle (UAV) classification.**

I. Introduction

With the widespread deployment of unmanned aerial vehicles (UAVs) in logistics, power inspection, forest fire monitoring, and urban security, a large number of small aerial targets have emerged in low-altitude airspace, posing potential threats to public safety and critical infrastructure. In particular, illegal UAV operations frequently occur around airports, high-speed railways, and sensitive areas, making it urgent to develop reliable techniques for UAV detection and classification [1].

Compared with optical or infrared approaches that are highly sensitive to illumination and weather conditions, radar offers all-weather capability, long-range detection, and high penetration, showing great promise in low-altitude surveillance and target recognition [2]. Extensive studies have demonstrated that radar echoes contain rich micro-Doppler signatures, such as rotor blade modulation frequency, modulation period, Doppler texture, and harmonic distribution, which effectively reflect the micromotion mechanism and structural characteristics of targets. These signatures serve as critical information sources for discriminating among UAVs. For instance, a micro-Doppler analysis framework was proposed in [3] by modeling target structural motion as modulation components in the radar spectrum, and it was applied to human activity recognition and UAV detection. Time–frequency analysis was later exploited for rotor modulation feature extraction [4], enabling discrimination of single-rotor and multirotor UAVs. Traditional methods often combine handcrafted features with classical classifiers, such as short-time Fourier transform (STFT)-based multiscale texture descriptors with support vector machines (SVMs) for UAV recognition [5]. On the contrary, deep learning enables end-to-end feature learning on time–frequency representations, effectively capturing harmonic structures and rotor periodicity for UAV classification and parameter estimation [6], [7], [8]. Recent works further integrate modulation texture and harmonic features with learning models to enhance small target recognition [9], [10].

In UAV target recognition tasks, feature selection plays a crucial role in enhancing classification performance and improving model generalization. Effective feature selection not only reduces redundant dimensions and alleviates the computational burden, but it also enables the model to focus on the most discriminative information. Existing studies on feature selection methods can be broadly divided into four categories.

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Guidong He and Peng Lei are with the School of Electronic and Information Engineering, Beihang University, Beijing 100191, China (e-mail: GuidongHe@buaa.edu.cn; buaaray@gmail.com).

Jun Wang is with Hangzhou Innovation Institute of Beihang University, Hangzhou 310051, China (e-mail: wangj203@buaa.edu.cn).

Jiangyou Zhu is with The Chinese University of Hong Kong, Hong Kong, China (e-mail: zj124@ie.cuhk.edu.hk).

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Filter methods evaluate each feature independently using statistical or information-theoretic criteria, such as the chi-square test, mutual information (MI), and correlation coefficients. However, they often neglect interfeature dependencies and may therefore fail to guarantee optimal discriminative power in complex scenarios [11], [12]. On the contrary, wrapper methods assess subsets of features by training and testing a classifier, with classification performance serving as the selection criterion. Techniques such as recursive feature elimination (RFE) and sequential forward/backward selection often achieve strong results but are computationally expensive [13], [14], [15]. To strike a balance between efficiency and accuracy, embedded methods integrate feature selection into model training. Examples include Lasso regression with L_1 regularization [16], and tree-based models such as random forests (RFs) and gradient boosting machines, which provide splitting-based importance measures [17], [18]. More recently, deep-learning-based approaches have introduced neural network–driven selection mechanisms, including sparsity-inducing constraints (e.g., L_1/L_0 regularization) [19], [20] and differentiable strategies such as the Gumbel-Softmax reparameterization [21], which enable the adaptive extraction of nonlinear and discriminative features from high-dimensional, complex signals.

Although existing methods have improved classification performance to some extent, most feature selection techniques are still based on a global criterion [22], i.e., that is, identifying a single optimal feature subset shared across all samples. However, in millimeter-wave radar UAV recognition tasks, micro-Doppler features often exhibit diverse patterns, nonstationarity, and multiscale characteristics. These features are strongly influenced by factors such as target type, flight state, rotor configuration, and radar observation angle, which leads to significant intersample variability. For example, even for the same type of UAV, the modulation texture can vary markedly under different pitch angles, while the micro-Doppler periodicity and spectral structures differ across flight modes such as hovering, forward flying, or accelerating. This cross-sample heterogeneity implies that not all features are equally effective or necessary for every instance—many features may become redundant or even misleading under specific conditions.

To address the above issue, this article proposes an instancewise feature selection [23] framework for radar UAV classification. In particular, the proposed method dynamically extracts task-relevant feature subsets for each sample, rather than relying on a globally consistent set of features. This design enables the model to better adapt to the heterogeneity of micro-Doppler signatures across different UAV types, flight states, and observation conditions. The main contributions of this work are summarized as follows.

1) *IB-guided instancewise feature selection framework:* We develop a new information-bottleneck (IB)-based framework that jointly performs instancewise feature selection and classification. Unlike global feature selection methods, the proposed framework learns an adaptive binary mask for each input sample, enabling personalized and data-dependent feature selection.
2) *Sparsity-aware prior for compact and discriminative feature subsets:* A sparsity-aware prior is incorporated into the IB formulation, encouraging the selector to retain only a compact set of highly discriminative features for each instance. This explicitly promotes sparse and interpretable selections, reduces redundancy, and enhances robustness.
3) *Efficient optimization via Gumbel-Softmax reparameterization:* To enable differentiable sampling of binary feature masks, the proposed framework employs the Gumbel-Softmax reparameterization within a variational inference formulation. This allows for stable end-to-end optimization of both the selector and the classifier, thus achieving efficient training and high-quality feature selection.

The rest of this article is organized as follows. Section II introduces the linear-frequency-modulated (LFM) radar echo model of the UAV and presents the extraction of micro-Doppler features. The details of the proposed instancewise feature selection classification network based on the IB principle are presented in Section III. Section IV reports the experimental results and performance evaluations. Finally, Section V concludes the article.

II. Radar Echo Modeling and Feature Extraction for UAVs

A. Radar Echo Model

Consider an LFM pulse radar system, where the transmitted signal is modeled as

$$s_t(t) = \exp\left(j\left(2\pi f_c t + \pi\mu t^2\right)\right), \quad 0 \le t \le T_p \tag{1}$$

with f_c denoting the carrier frequency, $\mu = B/T_p$ the chirp rate, B the signal bandwidth, and T_p the pulse duration. When a target is located at range R, the received echo signal can be expressed as

$$s_r(t) = \alpha \exp\left(j\left[2\pi f_c(t-\tau) + \pi\mu(t-\tau)^2\right]\right), \ 0 \le t \le T_p \tag{2}$$

where $\tau = 2R/c$ is the round-trip time delay, α is the reflection coefficient, and c is the speed of light. After matched filtering, the output corresponding to the target is given by

$$s(t) \approx \alpha \exp(j\phi)\ \mathrm{rect}\left(\tfrac{t-\tau}{T_p}\right) \tag{3}$$

where ϕ denotes a constant phase term caused by the time delay τ,and rect$(\cdot)$ represents the pulse envelope.

For UAV targets, the periodic motion of the rotor blades leads to an additional phase modulation

$$\Delta\phi(t) = \frac{4\pi}{\lambda}\Delta R(t) = \xi\cos(\omega_r t) \tag{4}$$

where $\Delta R(t) = A\cos(\omega_r t)$ represents the periodic displacement of the rotor blades, A is the micromotion amplitude induced by the rotor motion, ω_r is the angular frequency associated with the rotor rotation, and $\xi = 4\pi A/\lambda$ denotes the modulation index.

Accordingly, the echo from the nth rotor blade is expressed as

$$s_n(t) = \alpha_n \exp\left(j\left[\phi + \xi_n\cos(\omega_r t + \theta_n)\right]\right)\ \mathrm{rect}\left(\tfrac{t-\tau}{T_p}\right) \tag{5}$$

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TABLE I
EXTRACTED FEATURES FROM UAV TIME–FREQUENCY REPRESENTATIONS

Category	ID	Feature	Description
The Kim-Ling features [4]	F0	BW	Effective bandwidth estimated from the normalized average spectrum above a fixed threshold
	F1	σ_E/μ_E	Normalized standard deviation of total energy over time
Spectrogram statistics [24]	F2	μ_{spec}	Mean of the average spectrogram
	F3	σ_{spec}	Standard deviation of the average spectrogram
	F4	μ_{svd1}	Mean of the first left singular vector (SVD)
	F5	σ_{svd1}	Standard deviation of the first left singular vector
	F6	$\sigma_{\max}$	Maximum singular value of the spectrogram matrix
Cadence velocity diagram (CVD) [25]	F7	μ_{cvd}	Mean of CVD
	F8	σ_{cvd}	Standard deviation of CVD
Eigenvalue features [7]	F9-F13	$\{\lambda_1, \ldots, \lambda_5\}$	Top 5 eigenvalues of the frequency-domain correlation matrix
EMD-IMF features [26]	F14	ZC	Zero-crossing count
	F15	E_{norm}	Normalized energy ratio of each IMF
	F16	σ_{imf}	Standard deviation of IMF signal
	F17	$D_{max-mean}$	Normalized maximum-to-mean difference
	F18	H_E	Log-energy entropy
	F19	E_f	Total spectral energy density
	F20	$E_{f,\max}$	Maximum spectral energy density
	F21	Δf_{peak}	Normalized inter-peak frequency distance
Spectral kurtosis [27]	F22	μ_{SK}	Mean of spectral kurtosis
	F23	σ_{SK}	Standard deviation of spectral kurtosis
	F24	Skew_{SK}	Skewness of spectral kurtosis
	F25	Kurt_{SK}	Kurtosis of spectral kurtosis

where α_n is the scattering amplitude, $\xi_n = 4\pi A_n/\lambda$ denotes the modulation index, with A_n being the micromotion amplitude of the nth rotor blade, and θ_n represents the initial phase offset of the nth blade. The phase modulation gives rise to micro-Doppler sidebands spaced by ω_r in the spectrum.

Therefore, the complete UAV echo model can be formulated as the superposition of the body echo and the rotor-induced echoes

$$s_{\text{UAV}}(t) = s(t) + \sum_{n=1}^{N_s} s_n(t) \tag{6}$$

where N_s denotes the number of rotor blades. This formulation explicitly captures both the stable body scattering and the rotor-induced micromotion scattering, which together characterize the unique radar signatures of UAV targets.

B. Feature Extraction

Due to the micro-Doppler effect, UAV echo signals exhibit strong nonstationary characteristics. Time–frequency analysis is therefore commonly employed to represent their distinctive features.

The most widely used method is STFT, which is defined as

$$STFT(t, f) = \int s_{\text{UAV}}(\tau)\, w(\tau - t) \exp(-j2\pi f\tau)\, d\tau \tag{7}$$

where $w(\cdot)$ denotes a sliding window function. The spectrogram is obtained from the squared magnitude of the STFT, which provides an intuitive representation of the periodic micro-Doppler signatures induced by the rotor blade modulation.

Based on the spectrogram, a variety of discriminative features can be extracted, such as the Kim–Ling features, harmonic distribution, spectral texture descriptors, and eigenvalue features. These features capture both the structural characteristics of UAV rotors and their dynamic motion patterns, and they are widely utilized for classification and recognition tasks. A summary of the extracted features is provided in Table I.

III. PROPOSED FRAMEWORK

A. Problem Formulation

Let the extracted micro-Doppler feature matrix be denoted as $\mathbf{X} \in \mathbb{R}^{N\times D}$, where N is the number of micro-Doppler feature vectors and D is the dimensionality of each feature vector. The corresponding label vector is given by $\mathbf{Y} = [y_1, y_2, \ldots, y_N]^\top$, where each label satisfies $y_i \in \{0, 1, \ldots, c-1\}$ and c is the total number of UAV classes.

Since the micro-Doppler features in $\mathbf{X}$ are heterogeneous, a binary mask matrix $\mathbf{M} \in \mathbb{R}^{N\times D}$, $m_{ij} \in \{0, 1\}$, is introduced to enable instancewise feature selection, with the ith row $\mathbf{m}_i$ specifying the feature selection pattern for the corresponding sample $\mathbf{x}_i$. The masked feature representation is then given by the Hadamard product

$$\mathbf{Z} = \mathbf{M} \odot \mathbf{X} \tag{8}$$

where $\mathbf{z}_i = \mathbf{m}_i \odot \mathbf{x}_i$ represents the selected features of instance i.

The selected feature matrix $\mathbf{Z}$ is expected to be obtained such that high-quality features are preserved, noise is effectively compressed, and strong discriminative capability is maintained for UAV classification. In other words, the discriminative information of $\mathbf{Z}$ with respect to $\mathbf{Y}$ is retained, while noisy information inherited from $\mathbf{X}$ is discarded. To characterize these requirements, MI is introduced. In particular, $I(\mathbf{Z}; \mathbf{Y})$ is employed to measure the relevance between the selected representation and the class labels, while $I(\mathbf{X}; \mathbf{Z})$ quantifies the amount of information preserved from the input features [28].

Then, by jointly considering these two criteria, the UAV-classification problem is formulated as the following

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optimization problem:

$$\min \beta I(\mathbf{X};\mathbf{Z}) - I(\mathbf{Z};\mathbf{Y}) \tag{9}$$

where $I(\mathbf{X};\mathbf{Z})$ serves as the compression term, $I(\mathbf{Z};\mathbf{Y})$ acts as the prediction term, and $\beta > 0$ denotes a tradeoff hyperparameter balancing these two objectives. It is noteworthy that this optimization problem is consistent with the IB principle [29].

Moreover, it should be noted that the compression term $I(\mathbf{X};\mathbf{Z})$ in (9) does not explicitly induce sparsity in the feature space. In practice, information compression can be realized by feature transformations or low-rank projections, which do not guarantee that irrelevant feature dimensions are ignored. For radar-based micro-Doppler classification, however, it is desirable that the model explicitly identifies compact, instance-specific discriminative-feature subsets.

To this end, an additional sparsity constraint is imposed by introducing an L_1 penalty on the selection mask $\mathbf{M}$. This encourages each instance to activate only a small number of features. The final optimization problem is therefore given by

$$\min\ \beta I(\mathbf{X};\mathbf{Z}) - I(\mathbf{Z};\mathbf{Y}) + \eta \|\mathbf{M}\|_1 \tag{10}$$

where $\eta > 0$ is a regularization hyperparameter that controls the strength of the sparsity prior.

B. Solution Strategy

The optimization objective in (10) involves the MI terms $I(\mathbf{X};\mathbf{Z})$ and $I(\mathbf{Z};\mathbf{Y})$, which are generally intractable to compute or optimize directly, especially when $\mathbf{Z} = \mathbf{M} \odot \mathbf{X}$ is stochastic. To enable end-to-end training, we resort to variational inference, differentiable sampling, and lower bound approximations to relax and reformulate the objective into a differentiable training framework.

1) Compression Term: The compression term $I(\mathbf{X};\mathbf{Z})$ measures the amount of information retained in the intermediate representation $\mathbf{Z}$ about the input $\mathbf{X}$, which is defined as [30]

$$\begin{aligned} I(\mathbf{X};\mathbf{Z}) &= \iint p(\mathbf{x},\mathbf{z}) \log \frac{p(\mathbf{z}|\mathbf{x})}{p(\mathbf{z})}\, d\mathbf{x}\, d\mathbf{z} \\ &= \mathbb{E}_{p(\mathbf{x})}\left[D_{\mathrm{KL}}(p(\mathbf{z}|\mathbf{x}) \ \| p(\mathbf{z}))\right] \end{aligned} \tag{11}$$

where $p(\mathbf{x})$ is the marginal distribution of the input, $p(\mathbf{z}|\mathbf{x})$ is the conditional distribution of the representation given $\mathbf{x}$, $p(\mathbf{z}) = \int p(\mathbf{z}|\mathbf{x})p(\mathbf{x})\, d\mathbf{x}$ is the marginal distribution of $\mathbf{Z}$, and $p(\mathbf{x},\mathbf{z}) = p(\mathbf{x})p(\mathbf{z}|\mathbf{x})$ is the joint distribution. Here $D_{\mathrm{KL}}(\cdot\|\cdot)$ denotes the Kullback–Leibler (KL) divergence, and $\mathbb{E}_{p(\mathbf{x})}[\cdot]$ represents the expectation with respect to the distribution $p(\mathbf{x})$.

In the feature selection setting, the representation $\mathbf{Z}$ is obtained by applying a stochastic binary mask $\mathbf{M}$ to the input. Accordingly, $p(\mathbf{z}|\mathbf{x})$ can be equivalently expressed through the mask distribution $p(\mathbf{m}|\mathbf{x})$, and $p(\mathbf{z})$ is replaced by the corresponding prior distribution over the mask $p(\mathbf{m})$. Under this formulation, the compression term is approximated as

$$I(\mathbf{X};\mathbf{Z}) \approx \mathbb{E}_{p(\mathbf{x})}\left[D_{\mathrm{KL}}(q(\mathbf{m}|\mathbf{x}) \ \| p(\mathbf{m}))\right]. \tag{12}$$

Since both $p(\mathbf{m}|\mathbf{x})$ and $p(\mathbf{m})$ are unknown and generally intractable to compute in high-dimensional spaces, direct evaluation of $I(\mathbf{X};\mathbf{Z})$ is not feasible. To address the intractability of $p(\mathbf{m}|\mathbf{x})$, a variational distribution $q(\mathbf{m}|\mathbf{x})$ is introduced to approximate the true posterior. To make this variational distribution explicit, we need to specify how the mask vector $\mathbf{m}$ is generated. In the feature selection setting, each element of the mask vector $\mathbf{m}$ takes a binary value, indicating whether the corresponding feature is retained or discarded. Since this is inherently a binary decision, it is natural to model m_j as a Bernoulli random variable with selection probability $\pi_j(\mathbf{x})$. Assuming independence across features, the entire mask $\mathbf{m}$ follows a factorized Bernoulli distribution:

$$m_j \sim \text{Bernoulli}\left(\pi_j(\mathbf{x})\right), \quad j = 1, \ldots, D \tag{13}$$

where $\pi_j(\mathbf{x})$ denotes the selection probability of the jth feature for input $\mathbf{x}$.

Since sampling from the Bernoulli distribution is nondifferentiable and thus difficult to implement, we employ the Gumbel–Sigmoid reparameterization [21] to approximate the discrete sampling with a continuous and differentiable process

$$\begin{aligned} m_j &= \sigma\left(\frac{1}{\kappa}\left(\log \pi_j(\mathbf{x}) - \log\left(1 - \pi_j(\mathbf{x})\right) + g_j\right)\right) \\ g_j &\sim \text{Gumbel}(0,1) \end{aligned} \tag{14}$$

where $\sigma(\cdot)$ is the sigmoid function and κ is a temperature parameter controlling the smoothness of the approximation.

Thus, the variational distribution $q(\mathbf{m}|\mathbf{x})$ is explicitly defined through the Bernoulli mask with a differentiable relaxation. Formally, it takes the factorized Bernoulli form

$$q(\mathbf{m}|\mathbf{x}) = \prod_{j=1}^{D} \pi_j(\mathbf{x})^{m_j} \left(1 - \pi_j(\mathbf{x})\right)^{1-m_j}. \tag{15}$$

Having obtained the variational posterior $q(\mathbf{m}|\mathbf{x})$, we next turn to solving for the distribution $p(\mathbf{m})$, which is the true marginal distribution

$$p(\mathbf{m}) = \int p(\mathbf{m}|\mathbf{x})\ p(\mathbf{x})\ d\mathbf{x}$$

is generally intractable to compute in high-dimensional spaces.

To make the problem tractable, we replace the true marginal distribution with a simple alternative prior, which not only enables closed-form computation of the KL divergence term but also serves as a regularizer for the feature selection process. In particular, we still assume that the mask variables are independent and follow Bernoulli distributions with fixed parameters. This leads to the factorized prior

$$p(\mathbf{m}) = \prod_{j=1}^{D} p\left(m_j \mid \rho_j\right) = \prod_{j=1}^{D} \rho_j^{m_j} \left(1 - \rho_j\right)^{1-m_j} \tag{16}$$

where ρ_j is typically set to a constant, corresponding to a noninformative prior.

Note that the posterior $q(\mathbf{m}|\mathbf{x})$ and the prior $p(\mathbf{m})$ are both modeled as Bernoulli but parameterized differently. The posterior uses input-dependent parameters $\pi_j(\mathbf{x})$ to adaptively select features, while the prior adopts fixed parameters ρ_j to remain input-independent. If both used the same parameters, the KL divergence would collapse to zero and provide no regularization.

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After obtaining $q(\mathbf{m}|\mathbf{x})$ and $p(\mathbf{m})$, the KL divergence admits a closed-form expression

$$D_{\mathrm{KL}}(q(\mathbf{m}|\mathbf{x}) \,\|\, p(\mathbf{m})) = \sum_{j=1}^{D} \left[\pi_j(\mathbf{x}) \log \frac{\pi_j(\mathbf{x})}{\rho_j} + \left(1-\pi_j(\mathbf{x})\right) \log \frac{1-\pi_j(\mathbf{x})}{1-\rho_j} \right]. \tag{17}$$

Then, averaging over the dataset $\{\mathbf{x}_i\}_{i=1}^{N}$ gives the tractable approximation

$$I(\mathbf{X};\mathbf{Z}) \approx \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{D} \left[\pi_{ij} \log \frac{\pi_{ij}}{\rho_j} + \left(1-\pi_{ij}\right) \log \frac{1-\pi_{ij}}{1-\rho_j} \right] = \mathcal{L}_{\mathrm{KL}} \tag{18}$$

where $\pi_{ij} = \pi_j(\mathbf{x}_i)$ denotes the selection probability of the jth feature for the ith sample. This formulation depends only on the network outputs π_{ij}, leading to a differentiable and tractable feature selection mechanism that conforms to the IB principle.

2) Prediction Term: The prediction term $I(\mathbf{Z};\mathbf{Y})$ measures the amount of information retained in the intermediate representation $\mathbf{Z}$ about the target label $\mathbf{Y}$, defined as

$$\begin{aligned} I(\mathbf{Z};\mathbf{Y}) &= \iint p(\mathbf{z},y) \log \frac{p(y|\mathbf{z})}{p(y)} \, d\mathbf{z}\, dy \\ &= \mathbb{E}_{p(\mathbf{z},y)} \left[\log \frac{p(y|\mathbf{z})}{p(y)} \right] \qquad (19) \\ &= \mathbb{E}_{p(\mathbf{z},y)} \left[\log p(y|\mathbf{z}) \right] - \mathbb{E}_{p(y)} \left[\log p(y) \right] \\ &= \mathbb{E}_{p(\mathbf{z},y)} \left[\log p(y|\mathbf{z}) \right] + H(\mathbf{Y}) \qquad (20) \end{aligned}$$

where $p(\mathbf{z},y)$ denotes the joint distribution of the representation $\mathbf{Z}$ and the label y, and $p(y|\mathbf{z})$ is the conditional distribution of the scalar label y given the representation vector $\mathbf{z}$, $p(y)$ is the marginal distribution of y, $\mathbb{E}_{p(\mathbf{z},y)}[\cdot]$ represents the expectation with respect to the joint distribution $p(\mathbf{z},y)$, and $H(\mathbf{Y})$ is the entropy of the label distribution.

Similarly, the distribution of $\mathbf{Z}$ is intractable, which makes the direct computation of $I(\mathbf{Z};\mathbf{Y})$ difficult. To make it tractable, we adopt a variational lower bound by introducing a learnable distribution $q(y|\mathbf{z})$ to approximate the true posterior $p(y|\mathbf{z})$.

By introducing the variational distribution $q(y|\mathbf{z})$, we can decompose $\log p(y|\mathbf{z})$ into $\log q(y|\mathbf{z})$ plus a residual term

$$\mathbb{E}_{p(\mathbf{z},y)} \left[\log p(y|\mathbf{z}) \right] = \mathbb{E}_{p(\mathbf{z},y)} \left[\log q(y|\mathbf{z}) \right] + \mathbb{E}_{p(\mathbf{z})} \left[D_{\mathrm{KL}}(p(y|\mathbf{z}) \,\|\, q(y|\mathbf{z})) \right]. \tag{21}$$

Since the KL divergence is nonnegative

$$\mathbb{E}_{p(\mathbf{z},y)} \left[\log p(y|\mathbf{z}) \right] \geq \mathbb{E}_{p(\mathbf{z},y)} \left[\log q(y|\mathbf{z}) \right].$$

Therefore, the MI admits the following variational lower bound:

$$I(\mathbf{Z};\mathbf{Y}) \geq \mathbb{E}_{p(\mathbf{z},y)} \left[\log q(y|\mathbf{z}) \right] + H(\mathbf{Y}). \tag{22}$$

By ignoring $H(\mathbf{Y})$, which is a constant, and substituting $\mathbf{z} = \mathbf{m} \odot \mathbf{x}$, we obtain

$$\begin{aligned} I(\mathbf{Z};\mathbf{Y}) &\gtrsim \mathbb{E}_{p(\mathbf{z},y)} \left[\log q(y|\mathbf{z}) \right] \\ &= \int\int p(\mathbf{z},y) \, \log q(y|\mathbf{z}) \, d\mathbf{z}\, dy \\ &= \int\int p(\mathbf{x},y) \, q(\mathbf{m}|\mathbf{x}) \, \log q(y \mid \mathbf{m} \odot \mathbf{x}) \, d\mathbf{m}\, d\mathbf{x}\, dy \\ &= \mathbb{E}_{p(\mathbf{x},y)} \left[\mathbb{E}_{q(\mathbf{m}|\mathbf{x})} \log q(y \mid \mathbf{m} \odot \mathbf{x}) \right]. \qquad (23) \end{aligned}$$

To make the bound computable, the expectation over $p(\mathbf{x},y)$ is approximated by averaging over the data samples, and the stochastic mask $\mathbf{m}$ is sampled from $q(\mathbf{m}|\mathbf{x}_i)$ via the Gumbel–Sigmoid reparameterization. This yields

$$I(\mathbf{Z};\mathbf{Y}) \gtrsim \frac{1}{N} \sum_{i=1}^{N} \mathbb{E}_{\mathbf{m}\sim q(\mathbf{m}|\mathbf{x}_i)} \log q(y_i \mid \mathbf{m} \odot \mathbf{x}_i). \tag{24}$$

For each sample $(\mathbf{x}_i, y_i)$, the term $\log q(y_i \mid \mathbf{m} \odot \mathbf{x}_i)$ is the log-likelihood of the true label y_i under the predictive distribution $q(\cdot|\mathbf{z})$. Then, minimizing (10) requires maximizing the MI $I(\mathbf{Z};\mathbf{Y})$, which is equivalent to maximizing the expected log-likelihood of the true labels.

By taking the negative sign, the standard cross-entropy loss can be obtained

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{N} \sum_{i=1}^{N} \mathbb{E}_{\mathbf{m}\sim q(\mathbf{m}|\mathbf{x}_i)} \log q(y_i \mid \mathbf{m} \odot \mathbf{x}_i). \tag{25}$$

Therefore, maximizing $I(\mathbf{Z};\mathbf{Y})$ is equivalent to minimizing $\mathcal{L}_{\mathrm{CE}}$.

3) Implementation: Combining both the compression and prediction terms, the final objective function for instancewise feature selection is given by

$$\mathcal{L} = \beta \mathcal{L}_{\mathrm{KL}} + \mathcal{L}_{\mathrm{CE}} + \eta \|\mathbf{M}\|_1 \tag{26}$$

where the first term enforces information compression, the second term encourages predictive sufficiency, and the third term imposes explicit sparsity by penalizing the ℓ_1-norm of the feature selection mask $\mathbf{M}$. In particular, $\|\mathbf{M}\|_1 = \sum_{i=1}^{N} \sum_{j=1}^{D} |m_{ij}|$ measures the total number of active features across all samples, thus encouraging the model to select a minimal subset of discriminative features for each instance.

Fig. 1 illustrates the overall end-to-end neural network architecture that directly implements the derived objective. The architecture consists of two parts.

1) *Selector network:* It takes the input features $\mathbf{X}$ as input, learns the selection probabilities $\pi_j(\mathbf{x})$, and generates stochastic feature masks $\mathbf{M}$ via the Gumbel–Sigmoid reparameterization. This component corresponds to the compression term $I(\mathbf{X};\mathbf{Z})$, as it controls how much information from the input is retained in the masked representation $\mathbf{Z} = \mathbf{M} \odot \mathbf{X}$.

2) *Predictor network:* It receives the masked representation $\mathbf{Z}$ and performs label prediction through a classifier $q(y|\mathbf{z})$. This corresponds to the prediction term $I(\mathbf{Z};\mathbf{Y})$, as maximizing this MI is equivalent to maximizing the expected log-likelihood of the true labels.

The overall loss function combines these two objectives with the sparsity regularization term on the mask, and their relative contributions are balanced by hyperparameters.

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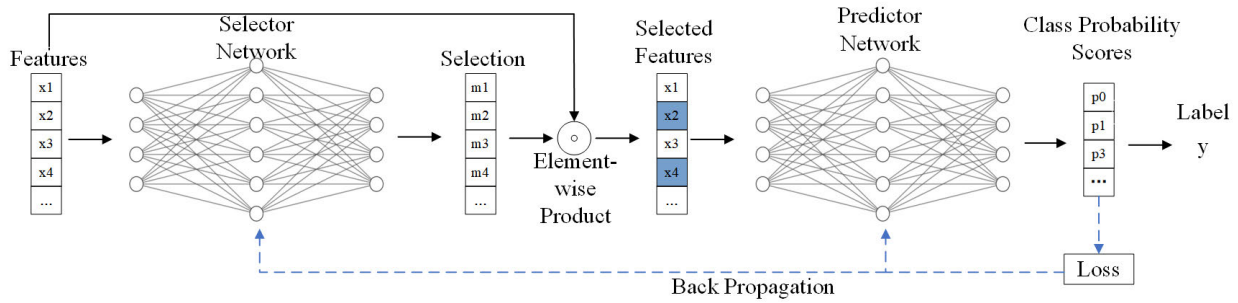


Fig. 1. End-to-end instancewise feature selection classification network architecture.



Fig. 2. Two types of UAV structures. (a) CW-10. (b) M300.

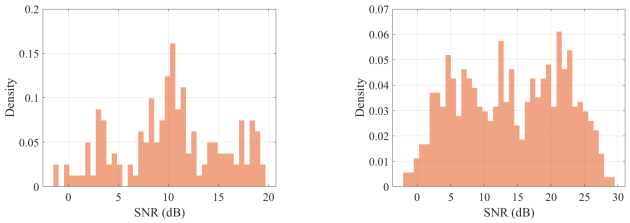


Fig. 3. SNR of two types of UAVs. (a) CW-10. (b) M300.

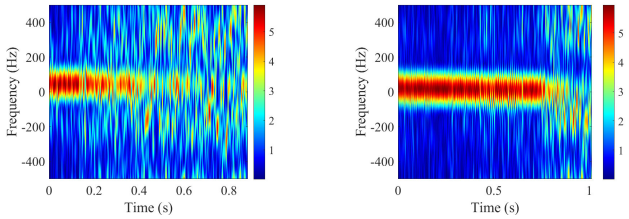


Fig. 4. Time–frequency spectrograms of UAV targets. (a) CW-10. (b) M300.

TABLE II
RADAR PARAMETERS

Parameter	Value	Unit
Carrier frequency	3.13344	GHz
Waveform	Linear frequency modulated pulse	–
Pulse repetition interval	1000	μs
IF sampling rate	491.52	Msps
Pulse width	2.0835	μs
Bandwidth	120	MHz

IV. EXPERIMENTS

A. Dataset and Radar Parameters

In this work, we employ a dataset collected by a ground-based radar under outdoor background conditions for the detection of low-altitude, slow-speed, and small-size (LSS) UAVs [31]. The dataset covers two types of UAV targets: the lightweight composite-wing CW-10 and the multirotor M300, as shown in Fig. 2. The CW-10 is a fixed-wing platform featuring a single propulsion motor and rigid aerodynamic surfaces, whereas the M300 is a quad-rotor system equipped with four high-speed rotating propellers and a central fuselage.

In the data acquisition process, the CW-10 was recorded under forward and backward flight trajectories, while the M300 measurements included hovering, straight inbound motion, and outbound S-shaped trajectories. However, before generating the time–frequency representations, the bulk translational motion of each UAV was compensated, ensuring that the remaining micro-Doppler signatures primarily reflected rotor-induced or structural vibrations rather than platform-level kinematics. The diverse flight states of the UAVs also lead to substantial variability in the signal-to-noise ratio (SNR) of their corresponding time–frequency representations. Fig. 3 illustrates the estimated SNR distribution obtained from the time–frequency spectrograms, where both UAV types exhibit SNR levels ranging approximately from 0 to 25 dB.

The data acquisition device is an S-band digital array radar experimental platform. Table II lists the radar parameters used in the experiments. Since the duration of a single measurement in the experiment is relatively long, a sliding-window technique is employed in order to segment the continuous data stream into shorter overlapping frames. Fig. 4 shows the time–frequency spectrograms of the CW-10 and M300 UAVs obtained from a single measurement, from which it can be observed that, due to various influencing factors, there exist unstable parts in the spectrograms, which inevitably lead to sample heterogeneity.

To ensure a fair and unbiased evaluation, we adopt a K-fold stratified cross-validation strategy. In particular, the entire dataset is randomly partitioned into $K = 5$ folds while maintaining approximately the same class distribution within each fold. In each iteration, one fold is used as the test set, and the remaining $K - 1$ folds are used for training. The process is repeated K times, and the average performance across all folds is reported as the final result. Table III summarizes the number of samples for the two UAV types in both the training and test sets used in the experiments.

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TABLE III
NUMBER OF SAMPLES FOR THE TWO UAV TYPES IN THE TRAINING AND TESTING SETS

Type	Training	Testing	Total
CW-10	1216	304	1520
M300	5472	1368	6840

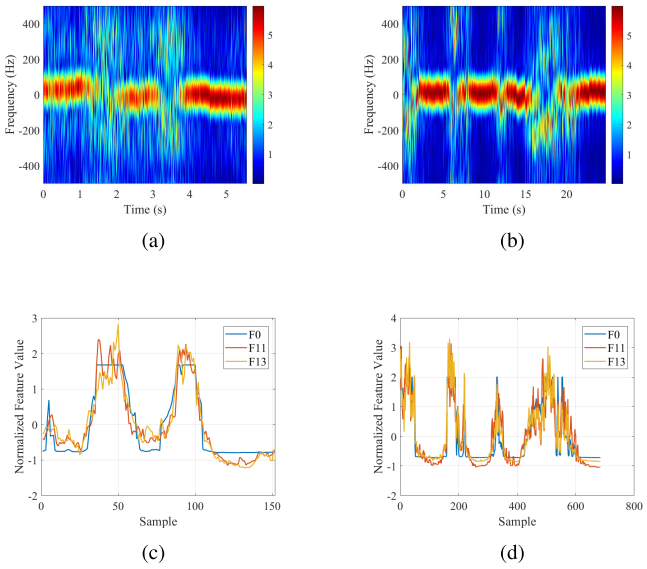


Fig. 5. Variation of features F0, F11, and F13 across samples of TFR for different UAVs. (a) TFR of CW-10. (b) TFR of M100. (c) Features of CW-10. (d) Features of M300.

B. Data Heterogeneity

In order to reveal the heterogeneity of UAV micro-Doppler features, this section presents a detailed analysis from two perspectives. On the one hand, the raw feature values are found to exhibit significant fluctuations across different samples, which can be attributed to the impact of interference, noise, and varying radar observation conditions. On the other hand, even for the same feature, its relative importance to the classification decision is not uniform across instances, reflecting instance-dependent contributions to the final prediction.

To facilitate the analysis, the two UAV categories were merged into a single dataset $\mathbf{X}$ with the binary labels $y \in \{0, 1\}$, where CW-10 was assigned label 0, and M300 label 1. Before further processing, all features were standardized to eliminate scale differences and ensure comparability.

First of all, taking F0, F11, and F13, which are listed in Table I as illustrative examples, their variations across samples are plotted in Fig. 5. It can be clearly observed that even within the same UAV type, the values of these features fluctuate considerably across different instances, revealing the instability induced by external interference and noise. In particular, samples with relatively large feature values often correspond to cases where the time–frequency signatures are damaged, where degraded SNRs lead to cluttered and less structured time–frequency distributions. Such fluctuations highlight the first aspect of feature heterogeneity, namely that the same feature may not remain stable or reliable across all instances of the same class.

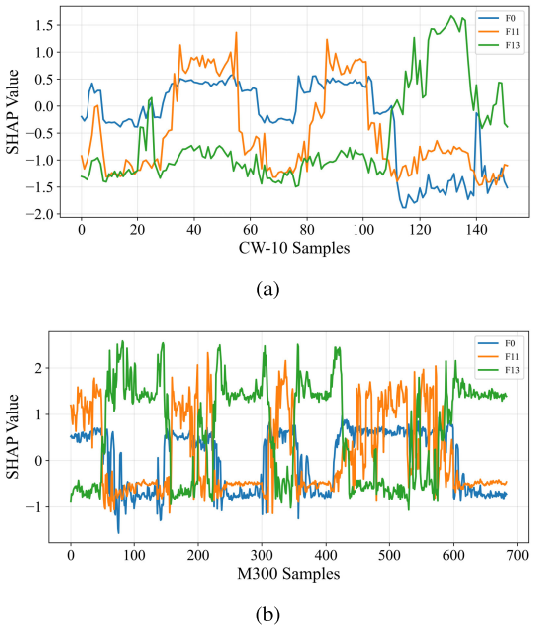


Fig. 6. SHAP values of features F0, F11, and F13 across samples for different UAVs. (a) CW-10. (b) M300.

In addition, to further highlight the second aspect of feature heterogeneity, we investigated how the same feature contributes differently to classification across individual samples. To this end, Shapley Additive exPlanations (SHAP) [32] values were employed as an instancewise attribution method, which provides a unified and theoretically grounded measure of the marginal contribution of each feature to the prediction outcome of each sample. Unlike global feature importance metrics that only capture averaged effects, SHAP enables a fine-grained characterization of feature relevance at the instance level.

Accordingly, we computed and visualized the SHAP values of features F0, F11, and F13 for both UAV categories, as illustrated in Fig. 6. It can be observed that the SHAP values fluctuate substantially across samples, even for the same feature. This indicates that identical features may support classification positively for certain instances while playing a negligible or even adverse role for others. Such variations reveal that the discriminative power of features is highly instance-dependent, which further substantiates the heterogeneity of UAV micro-Doppler features.

C. Analysis of Selected Features

To further illustrate the effectiveness of the proposed instancewise feature selection method, we provide visualization results on test samples. We randomly selected 20 test samples, and Fig. 7 illustrates their standardized feature values across all feature dimensions. Fig. 8 presents the corresponding feature selection masks, indicating whether each feature dimension is selected by the model for a given instance. Dark shading denotes selected features, while light shading indicates unselected ones. The selected features corresponding to Figs. 7 and 8 are shown in Fig. 9, where redundant dimensions are suppressed, and only discriminative features are retained. It can be observed that some features demonstrate strong discriminative patterns across different samples, while the

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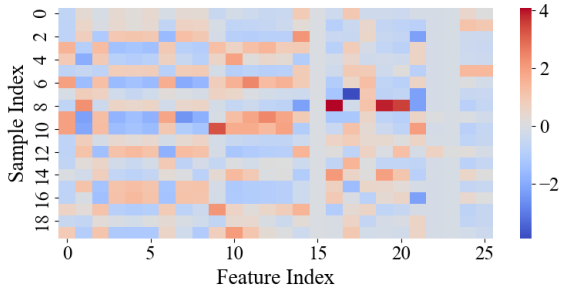


Fig. 7. Standardized feature distributions of 20 test samples.

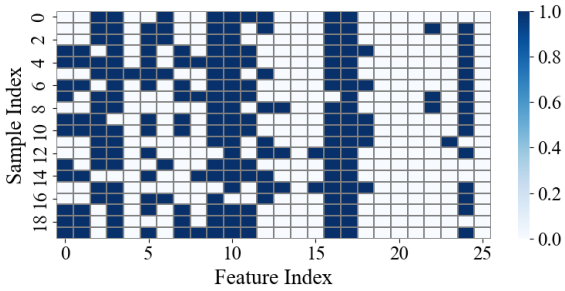


Fig. 8. Feature selection masks of 20 test samples. Darker colors correspond to features retained by the selector, while lighter colors indicate features not selected.

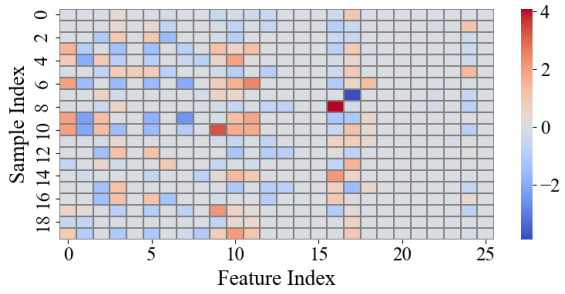


Fig. 9. Selected features of 20 test samples.

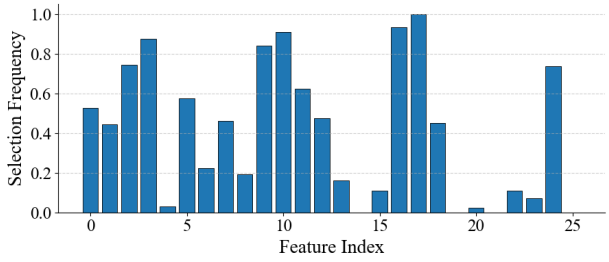


Fig. 10. Average selection frequency distribution of all test features.

masks effectively filter out irrelevant dimensions, resulting in sparse yet highly informative representations.

Furthermore, Fig. 10 depicts the overall feature selection frequency distribution. It can be observed that only a small subset of features is consistently selected with high frequency, while the majority of dimensions are rarely chosen. This indicates that the proposed approach is able to automatically focus on the most discriminative feature dimensions, effectively achieving both dimensionality reduction and feature extraction. Moreover, the frequency with which features are

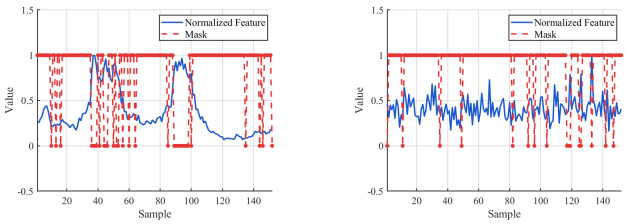


Fig. 11. Two discriminative features and corresponding selection masks. (a) F10. (b) F17.

selected is not uniform, suggesting that the key discriminative features vary across different samples.

Fig. 11 visualizes examples of the time–frequency representations of the CW-10, together with two discriminative features, F10 and F17, and their corresponding selection masks. Feature F10, which exhibits noticeable fluctuations as the quality of the time–frequency representation varies, reflects instability across samples. Accordingly, our method tends to suppress this feature when its value becomes unreliable. On the contrary, feature F17 remains largely invariant to changes in the time–frequency representation and demonstrates strong discriminability with high consistency. Thus, the proposed selector tends to preserve this feature across different samples.

D. Classification Performance Comparison

In this section, we first compare several representative traditional classification methods, including logistic regression (LR) [33], SVM [34], RF [17], and a baseline multilayer perceptron (MLP) [35]. All models are evaluated under the same data partitioning strategy, and their results serve as reference baselines for classification performance.

To further assess the effectiveness of feature selection strategies, we conduct experiments by integrating MLP classifiers with different feature selection techniques, including MI [36], chi-square test [37], RFE [14], Lasso regularization [16], and RF feature importance (RF Importance) [38]. This design enables us to directly compare the performance between using raw features and applying feature selection before classification, thereby validating the advantages of the proposed instancewise feature selection approach. Considering the widespread use of deep learning models in micro-Doppler-based target recognition tasks, three representative architectures, including a CNN [39], ResNet18 [40], and MobileNetV2 [41], were additionally implemented for a comprehensive comparison. Unlike traditional machine-learning approaches that operate on handcrafted micromotion features, these deep learning models directly ingest the time–frequency representations.

For performance evaluation, we adopt three widely used metrics: classification accuracy (Acc), the area under the receiver operating characteristic curve (AUC), and the average precision (APR). All results are reported as the mean and standard deviation over $K = 5$-fold stratified cross-validation. The experimental results are summarized in Table IV. Linear baselines such as LR and SVM achieve limited performance, while nonlinear models (RF and MLP) show clear improvements.

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TABLE IV
CLASSIFICATION PERFORMANCE COMPARISON OF DIFFERENT METHODS

Method	Accuracy	AUC	APR
Baseline Methods			
Logistic Regression [33]	0.880 ± 0.025	0.908 ± 0.021	0.975 ± 0.006
SVM [34]	0.880 ± 0.025	0.908 ± 0.021	0.975 ± 0.006
Random Forest [17]	0.894 ± 0.012	0.909 ± 0.010	0.969 ± 0.003
MLP [35]	0.901 ± 0.021	0.936 ± 0.020	0.981 ± 0.010
Feature Selection + MLP			
Filter - Mutual Info [36]	0.865 ± 0.023	0.885 ± 0.033	0.972 ± 0.009
Filter - Chi2 [37]	0.861 ± 0.017	0.891 ± 0.011	0.974 ± 0.003
Wrapper - RFE [14]	0.887 ± 0.012	0.936 ± 0.008	0.985 ± 0.001
Embedded - Lasso [16]	0.900 ± 0.030	0.931 ± 0.028	0.980 ± 0.010
Embedded - RF Importance [38]	0.890 ± 0.020	0.931 ± 0.029	0.984 ± 0.007
Deep Learning Model			
CNN [39]	0.896 ± 0.037	0.953 ± 0.021	0.989 ± 0.005
ResNet18 [40]	0.907 ± 0.080	0.996 ± 0.006	0.999 ± 0.001
MobileNetV2 [41]	0.904 ± 0.048	0.991 ± 0.006	0.998 ± 0.001
Ours	**0.913 ± 0.021**	0.949 ± 0.019	0.988 ± 0.005

TABLE V
ABLATION RESULTS UNDER DIFFERENT LOSS TERM SETTINGS

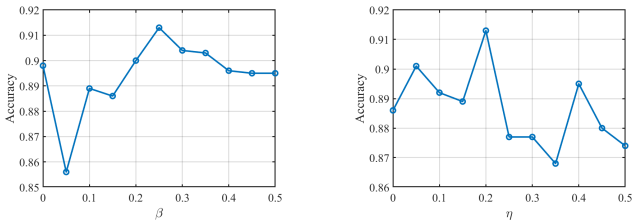
Component Setting	Accuracy	AUC	APR
CE	0.874±0.025	0.897±0.028	0.970±0.015
CE + Sparsity	0.898±0.010	0.945±0.008	0.987±0.002
KL + CE	0.886±0.010	0.933±0.019	0.985±0.005
KL + CE + Sparsity	**0.913±0.021**	**0.949±0.019**	**0.988±0.005**

However, conventional feature selection strategies, including filter, wrapper, and embedded-based methods, do not lead to further gains and in some cases even degrade performance. As for deep learning models, they tend to capture global structures from noisy spectrograms and require significantly larger training sets to generalize reliably. As a result, they exhibit noticeable performance variability and do not consistently surpass traditional classifiers. On the contrary, our proposed instancewise feature selection approach achieves the best overall results, reaching an accuracy of 0.913, an AUC of 0.949, and an APR of 0.988. These results confirm that the proposed method can adaptively select the most discriminative features for each instance, thereby improving classification accuracy while enhancing interpretability.

E. Experiment Analysis

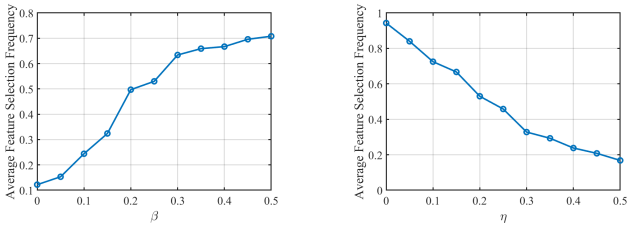
To assess the individual contribution of each component in the proposed objective, namely the KL divergence term, the cross-entropy term, and the sparsity regularization term, a series of ablation experiments were conducted. Table V summarizes the classification performance under these different loss configurations. The results demonstrate that removing any single component leads to a noticeable degradation in accuracy, AUC, and APR, confirming that all three terms play essential and complementary roles in the effectiveness of the proposed IFS framework.

Furthermore, to investigate the influence of the hyperparameters β and η, we conduct a comprehensive experiment to examine their impact on model performance. In particular, due to the three loss components operating at a comparable scale and the cross-entropy term is dominant, we vary only one hyperparameter at a time, either β or η, in the range [0, 0.5] with a step size of 0.05. In each experiment, we evaluate the resulting classification accuracy, the average feature selection frequency, and the divergence between the selected feature subset and the original feature distribution while keeping the other hyperparameter fixed. Figs. 12–14 summarize the results. Fig. 12 shows that optimal performance requires a proper balance between β and η, revealing a coupled dependence rather than a monotonic trend—indicating that KL regularization and



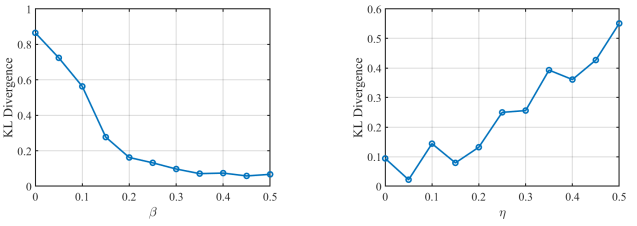
(a) (b)

Fig. 12. Variation of classification accuracy with the β and η. (a) Fixed η = 0.2. (b) Fixed β = 0.25.



(a) (b)

Fig. 13. Variation of the average feature selection frequency with the β and η. (a) Fixed η = 0.2. (b) Fixed β = 0.25.



(a) (b)

Fig. 14. Variation of the divergence between selected features and the original features with β and η. (a) Fixed η = 0.2. (b) Fixed β = 0.25.

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TABLE VI
PERFORMANCE UNDER DIFFERENT K-FOLD SETTINGS

K	Accuracy	AUC	APR
2	0.900 ± 0.000	0.948 ± 0.011	0.988 ± 0.003
3	0.903 ± 0.013	0.934 ± 0.021	0.983 ± 0.006
4	0.915 ± 0.013	0.952 ± 0.006	0.988 ± 0.002
5	0.913 ± 0.021	0.949 ± 0.019	0.988 ± 0.005

sparsity jointly shape the behavior of the instancewise feature selection module. As observed in Fig. 13, η predominantly controls the number of selected features, with increasing η leading to stronger sparsity. On the contrary, larger β tends to raise the selection rate because stronger KL regularization encourages the selected subset to stay closer to the original feature distribution. Fig. 14 further shows that larger β results in smaller divergence, meaning the selected features better preserve the structure of the full feature vector. Conversely, increasing η enforces stricter sparsity and therefore increases divergence, as fewer selected features provide a less complete approximation of the original representation.

In addition, to evaluate the impact of data partitioning on performance stability, we performed additional experiments under multiple K-fold configurations, including $K = 2, 3, 4$, and 5. As summarized in Table VI, the obtained accuracy, AUC, and APR values exhibit only minor fluctuations across different K values, indicating that the proposed method is robust with respect to the choice of K and that the reported results are not an artifact of a particular fold setting.

Then, in K = fivefold, we further evaluate the computational complexity of the proposed method. The proposed framework contains a feature selector and a shallow classifier, both implemented as lightweight fully connected networks operating on the D = 26-D handcrafted feature vector. The selector is a two-layer MLP (128–128) with a sigmoid output of size 26 to produce instancewise selection probabilities. The classifier is a 256–256 MLP with dropout, followed by a two-unit softmax layer for binary prediction. For a single forward pass, the computational cost is dominated by matrix multiplications in both modules, yielding a total complexity of

$$\mathcal{O}\left(DH_{\text{sel}} + H_{\text{sel}}^2 + DH_{\text{cls}} + H_{\text{cls}}^2\right)$$

where H_{sel} and H_{cls} denote the hidden-layer widths of the selector and classifier. To evaluate efficiency, we benchmarked the model on a 12th-Gen Intel i7-12700H CPU. Training requires about 10.36 s per fold, while inference achieves a latency of 0.0087 ms per sample, indicating real-time capability. The overall model together contains only about 96-k parameters, illustrating the lightweight nature of our framework.

At last, to evaluate the proposed method on a more challenging dataset with a larger variety of UAV types, we adopt the publicly released LSS-FMCWR-2.0 dataset [42], which contains four rotorcraft UAV platforms, including DJI M350, DJI Inspire 2, DJI Mavic 2, and a hexacopter. The dataset was collected using a 23.7-GHz FMCW radar. Table VII summarizes the classification performance of different methods on the LSS-FMCWR-2.0 dataset. As shown, the proposed method consistently maintains a noticeable performance advantage over both traditional machine-learning methods and deep learning models, even under the more diverse and challenging multi-UAV scenario.

TABLE VII
CLASSIFICATION PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE LSS-FMCWR-2.0 DATASET

Method	Accuracy	AUC	APR
Baseline Methods			
Logistic Regression	0.707	0.903	0.790
SVM	0.717	0.940	0.851
Random Forest	0.770	0.948	0.872
MLP	0.755	0.941	0.856
Feature Selection + MLP			
Filter – Mutual Info	0.754	0.930	0.829
Filter – Chi2	0.750	0.922	0.807
Wrapper – RFE	0.787	0.947	0.863
Embedded – Lasso	0.755	0.941	0.856
Embedded – RF Importance	0.764	0.926	0.820
Deep Learning Models			
CNN	0.759	0.913	0.806
ResNet18	0.777	0.935	0.845
MobileNetV2	0.782	0.959	0.898
Ours	**0.841**	**0.971**	**0.913**

V. CONCLUSION

This article proposed an instancewise feature selection framework for UAV radar classification, motivated by the heterogeneity of micro-Doppler signatures across UAV categories, flight states, and observation conditions. By formulating the task within the IB principle and incorporating a sparsity prior, we developed an end-to-end network that dynamically selects discriminative features for each sample. The experimental results on real radar datasets demonstrated that the proposed method not only improves classification accuracy over conventional baselines and global feature selection strategies but also enhances interpretability by producing feature masks that align with physically meaningful micro-Doppler patterns.

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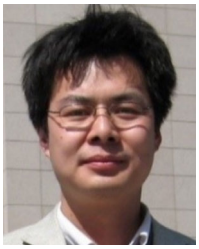
Guidong He (Graduate Student Member, IEEE) received the B.S. degree in electrical engineering from Beihang University, Beijing, China, in 2021. He is currently pursuing the Ph.D. degree in electronics and information engineering with the School of Electronic and Information Engineering, Beihang University.

His research interests include signal processing and target recognition.



Peng Lei received the B.S. and Ph.D. degrees in electrical engineering from Beihang University, Beijing, China, in 2006 and 2012, respectively.

He is currently an Associate Professor with the School of Electronic and Information Engineering, Beihang University. His research interests include signal processing, especially in time–frequency analysis and spectral estimation, image processing, target recognition, and machine learning in electrical engineering.



Jun Wang received the B.S. degree in electronic and information engineering from Northwest Polytechnic University, Xi'an, China, in 1995, and the M.S. and Ph.D. degrees in electronic and information engineering from Beihang University, Beijing, China, in 1998 and 2001, respectively.

He is currently a Professor with the School of Electronic and Information Engineering, and also the Dean of Hangzhou Innovation Institute of Beihang University, Hangzhou, China. His research interests include signal processing, DSP/FPGA real-time architecture, and target recognition and tracking.



Jiangyou Zhu received the M.S. degree with the School of Electronic and Information Engineering, Beihang University, Beijing, China, in 2021. She is currently pursuing the Ph.D. degree with the Department of Information Engineering, The Chinese University of Hong Kong, Hong Kong, China.

Her research interests include radar signal processing, especially in DOA disambiguation and target recognition.

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